Advanced HTML and CSS

Become the CSS expert you deserve to be — build beautiful, accessible, responsive websites



Course Overview

- Dig deeper into CSS and become a true expert
- Build web projects that look beautiful on any device
- Three major focus areas:
  - Animation
  - Accessibility
  - Responsive Design
- Ends with a hands-on Homepage project

Animation: Transforms & Transitions

- Transforms — move, rotate, scale, and skew elements
  - translate, rotate, scale, skew
- Transitions — smooth changes between states
  - transition-property, transition-duration, transition-timing-function
- Used for hover effects, button feedback, and subtle motion

Animation: Keyframes

- @keyframes — define multi-step animations
- Control intermediate states with percentages (0%, 50%, 100%)
- Animation properties — name, duration, iteration-count, direction, fill-mode
- Use for loading spinners, attention grabbers, and complex sequences

Accessibility: Introduction & WCAG

- Introduction to Web Accessibility — building for everyone
- The Web Content Accessibility Guidelines (WCAG) — the global standard
- Four core principles (POUR):
  - Perceivable, Operable, Understandable, Robust
- Accessibility is not optional — it's part of quality

Accessibility: Semantic HTML & Colors

- Semantic HTML — use elements for their meaning (header, nav, main, footer)
- Benefits: better structure, SEO, and screen reader support
- Accessible Colors — sufficient contrast between text and background
- Don't rely on color alone to convey information

Accessibility: Navigation, Text & ARIA

- Keyboard Navigation — every interactive element must be reachable and usable via keyboard
- Meaningful Text — descriptive link text and alt text for images
- WAI-ARIA — attributes to enhance accessibility when HTML alone isn't enough
  - roles, states, properties
- Accessibility Auditing — test with tools like Lighthouse and screen readers

Responsive Design: Introduction & Natural Responsiveness

- Introduction to Responsive Design — one site, every screen size
- Natural Responsiveness — start with flexible, fluid layouts
- Avoid fixed widths; use percentages and relative units
- Mobile-first thinking

Responsive Design: Images & Media Queries

- Responsive Images — serve the right image for the right screen
  - srcset, sizes, and the picture element
- Media Queries — apply CSS only at certain screen sizes
  - @media (min-width: 768px) { … }
- Common breakpoints: mobile, tablet, desktop

Homepage & Key Takeaways

- **Homepage:**

- Build a complete, responsive homepage from scratch
- Apply animation, accessibility, and responsive techniques together

- **Key takeaways:**

- Animate with transforms, transitions, and keyframes
- Build accessible sites with semantic HTML, good contrast, and ARIA
- Design responsively with fluid layouts, responsive images, and media queries
- Test with auditing tools before you ship